

Problem 1

(a) Let x_1 and x_2 be the numbers of mugs and bowls produced. The LP is

$$\max 30x_1 + 40x_2 \quad \text{subject to} \quad \begin{cases} x_1 + 2x_2 \leq 80 & (\text{clay}), \\ 3x_1 + x_2 \leq 90 & (\text{glaze}), \\ x_1, x_2 \geq 0. \end{cases}$$

(b) Introduce unused clay $s_1 \geq 0$ and unused glaze $s_2 \geq 0$. The objective is unchanged, and

$$\begin{cases} x_1 + 2x_2 + s_1 = 80, \\ 3x_1 + x_2 + s_2 = 90, \\ x_1, x_2, s_1, s_2 \geq 0. \end{cases} \quad A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 3 & 1 & 0 & 1 \end{pmatrix}.$$

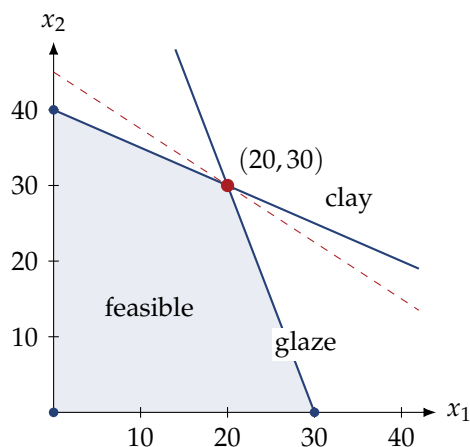
Thus $n = 4$, $m = 2$. The last two columns form I_2 , so $\text{rank}(A) = 2$.

(c) The four columns are nonzero and no two are proportional, so all six pairs are independent. Setting the nonbasic variables to zero gives:

Basis	x_1	x_2	s_1	s_2	Feasible?	Margin (DH)
$\{x_1, x_2\}$	20	30	0	0	Yes	1800
$\{x_1, s_1\}$	30	0	50	0	Yes	900
$\{x_1, s_2\}$	80	0	0	-150	No	—
$\{x_2, s_1\}$	0	90	-100	0	No	—
$\{x_2, s_2\}$	0	40	0	50	Yes	1600
$\{s_1, s_2\}$	0	0	80	90	Yes	0

For example, the basis $\{x_1, x_2\}$ gives $x_1 + 2x_2 = 80$ and $3x_1 + x_2 = 90$, hence $(x_1, x_2) = (20, 30)$.

(d) The feasible polygon has vertices $(0, 0)$, $(30, 0)$, $(20, 30)$, and $(0, 40)$. Their lifts are exactly the four BFSs in the table.



The dashed line is

$$30x_1 + 40x_2 = 1800.$$

It touches the feasible polygon only at $(20, 30)$. Comparing the four vertex values therefore gives

$$x_1^* = 20, \quad x_2^* = 30,$$

$$z^* = 1800 \text{ DH.}$$

Problem 2

(a) Let $x_1, x_2 \geq 0$ be the kilograms of barley meal and alfalfa pellets per cow per day. The LP is

$$\min 5x_1 + 8x_2 \quad \text{subject to} \quad \begin{cases} 2x_1 + x_2 \geq 8 & (\text{protein}), \\ x_1 + 3x_2 \geq 9 & (\text{fibre}), \\ x_1, x_2 \geq 0. \end{cases}$$

(b) Negate the objective and subtract nonnegative surplus variables u_1, u_2 :

$$\begin{array}{ll} \max & -5x_1 - 8x_2 \\ \text{s.t.} & 2x_1 + x_2 - u_1 = 8, \\ & x_1 + 3x_2 - u_2 = 9, \\ & x_1, x_2, u_1, u_2 \geq 0. \end{array}$$

Here $u_1 = 2x_1 + x_2 - 8$ is protein supplied above the minimum, and $u_2 = x_1 + 3x_2 - 9$ is excess fibre. An ordinary added slack would encode a $\leq$ constraint; these $\geq$ constraints require a subtracted surplus.

(c) For the basis $\{u_1, u_2\}$, set $x_1 = x_2 = 0$. Then

$$-u_1 = 8, \quad -u_2 = 9, \quad (x_1, x_2, u_1, u_2) = (0, 0, -8, -9).$$

The basis matrix is $-I_2$, which is invertible, but the basic solution is infeasible. A method requiring an initial BFS cannot start from this basis; one must first find a feasible basis.

(d) The matrix is

$$A = \begin{pmatrix} 2 & 1 & -1 & 0 \\ 1 & 3 & 0 & -1 \end{pmatrix}.$$

Every pair of columns is independent. The six basic solutions are:

Basis	x_1	x_2	u_1	u_2	Feasible?	Cost (DH)
$\{x_1, x_2\}$	3	2	0	0	Yes	31
$\{x_1, u_1\}$	9	0	10	0	Yes	45
$\{x_1, u_2\}$	4	0	0	-5	No	—
$\{x_2, u_1\}$	0	3	-5	0	No	—
$\{x_2, u_2\}$	0	8	0	15	Yes	64
$\{u_1, u_2\}$	0	0	-8	-9	No	—

The cost is bounded below by zero and feasible solutions exist, so an optimal BFS exists. The smallest of the three feasible basic costs is 31:

$$x_1^* = 3 \text{ kg}, \quad x_2^* = 2 \text{ kg}, \quad C^* = 31 \text{ DH.}$$

This ration supplies $2(3) + 2 = 8$ protein units and $3 + 3(2) = 9$ fibre units, meeting both minima exactly.

Problem 3

(a) Let x_1, x_2, x_3 be the hectares assigned to wheat, sunflower, and fallow. In thousands of DH, the LP is

$$\begin{array}{ll}\max & 4x_1 + 5x_2 + x_3 \\ \text{s.t.} & x_1 + x_2 + x_3 = 100, \\ & 3x_1 + 2x_2 \leq 240, \\ & x_1 \geq 20, \quad x_2, x_3 \geq 0.\end{array}$$

The land constraint is an equality because every hectare must receive one of the three uses; unplanted land is explicitly counted as fallow.

(b) Set $u = x_1 - 20 \geq 0$, so $x_1 = 20 + u$. Adding a water slack $s \geq 0$ gives

$$\begin{array}{ll}\max & 80 + 4u + 5x_2 + x_3 \\ \text{s.t.} & u + x_2 + x_3 = 80, \\ & 3u + 2x_2 + s = 180, \\ & u, x_2, x_3, s \geq 0.\end{array}$$

The constant 80 may be omitted when finding the optimizer, but must be included when reporting the margin. With variable order (u, x_2, x_3, s) ,

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 80 \\ 180 \end{pmatrix}.$$

The last two columns form I_2 , so the two rows are independent.

(c) All six column pairs are independent. Their basic solutions are:

Basis	u	x_2	x_3	s	Feasible?	Margin (10^3 DH)
$\{u, x_2\}$	20	60	0	0	Yes	460
$\{u, x_3\}$	60	0	20	0	Yes	340
$\{u, s\}$	80	0	0	-60	No	—
$\{x_2, x_3\}$	0	90	-10	0	No	—
$\{x_2, s\}$	0	80	0	20	Yes	480
$\{x_3, s\}$	0	0	80	180	Yes	160

Thus there are four BFSs. The largest objective value is 480.

(d) The optimal transformed solution is $(u, x_2, x_3, s) = (0, 80, 0, 20)$. Returning to the original variables,

$$(x_1^*, x_2^*, x_3^*) = (20, 80, 0), \quad \text{margin} = 480000 \text{ DH.}$$

All 100 hectares are assigned, and the minimum-wheat requirement is tight. Water use is

$$3(20) + 2(80) = 220 < 240,$$

so the water constraint is not binding: 20 units remain unused. The nonnegativity constraint on fallow land is also tight.

Problem 4

(a) Let x_1, x_2, x_3 be the numbers of posters assigned to presses A, B, and C. Their running times are $x_1/500$, $x_2/300$, and $x_3/200$ hours. Their maximum is piecewise linear, not a single linear objective. Introduce t bounding all three times:

$$\begin{array}{ll} \min & t \\ \text{s.t.} & x_1 + x_2 + x_3 = 6000, \\ & x_1 \leq 500t, \quad x_2 \leq 300t, \quad x_3 \leq 200t, \\ & x_3 \leq 1000, \\ & x_1, x_2, x_3, t \geq 0. \end{array}$$

For any allocation, minimizing t sets it to the largest running time.

(b) Add slacks s_1, s_2, s_3 for the time constraints and s_4 for the ink constraint. The equational LP minimizes t subject to

$$\begin{array}{ll} x_1 + x_2 + x_3 = 6000, \\ x_1 - 500t + s_1 = 0, \\ x_2 - 300t + s_2 = 0, & x_1, x_2, x_3, t, s_1, s_2, s_3, s_4 \geq 0. \\ x_3 - 200t + s_3 = 0, \\ x_3 + s_4 = 1000, \end{array}$$

Hence $n = 8$ and $m = 5$. Equivalently, one may maximize $-t$.

(c) The proposed quantities sum to 6000, and their running times are 6.25, 6.25, and 5 hours. Thus the plan is feasible with

$$(s_1, s_2, s_3, s_4) = (0, 0, 250, 0).$$

Choose $B = \{x_1, x_2, x_3, t, s_3\}$. In the equation order above, its five columns form

$$A_B = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & -500 & 0 \\ 0 & 1 & 0 & -300 & 0 \\ 0 & 0 & 1 & -200 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

If $A_B a = 0$, the last row gives $a_3 = 0$, and rows 2 and 3 give $a_1 = 500a_4$, $a_2 = 300a_4$. Row 1 then gives $800a_4 = 0$. Hence $a_1 = a_2 = a_3 = a_4 = 0$, and row 4 gives $a_5 = 0$. The columns are independent.

The three nonbasic slacks are zero, and the five basic values are 3125, 1875, 1000, 25/4, 250, all strictly positive. This is a **nondegenerate BFS**.

(d) With unlimited ink, $6000 \leq (500 + 300 + 200)t$ gives $t \geq 6$. The allocation (3000, 1800, 1200) attains this bound, so the optimum is **6 hours**. With limited ink,

$$6000 = x_1 + x_2 + x_3 \leq 800t + 1000 \implies t \geq \frac{25}{4} = 6.25.$$

The plan in (c) attains this bound. The ink restriction therefore adds 0.25 hour, i.e. **15 minutes**, to the optimal completion time.

Problem 5

(a) Index March, April, and May by $t = 1, 2, 3$, and set $d = (100, 180, 120)$. Let p_t be production and I_t be inventory minus backlog at the end of month t . Then

$$I_t = I_{t-1} + p_t - d_t \quad (t = 1, 2, 3), \quad I_0 = 0, \quad I_3 = 0,$$

with $0 \leq p_t \leq 150$ and $I_1, I_2 \in \mathbb{R}$. Explicitly,

$$I_1 = p_1 - 100, \quad I_2 = I_1 + p_2 - 180, \quad I_3 = I_2 + p_3 - 120 = 0.$$

Writing $a^+ = \max\{a, 0\}$, the total cost to minimize is

$$C = 3000 \sum_{t=1}^3 p_t + 200 \sum_{t=1}^3 (I_t)^+ + 500 \sum_{t=1}^3 (-I_t)^+.$$

The last two sums charge inventory or backlog carried at each month end. Their May terms are zero because $I_3 = 0$.

(b) Substitute $I_t = h_t - q_t$, where $h_t, q_t \geq 0$, for each $t = 1, 2, 3$. Introduce unused production capacities $s_t \geq 0$. Keeping the terminal balance as an equation, the equational model is

$$\min C = 3000(p_1 + p_2 + p_3) + 200(h_1 + h_2 + h_3) + 500(q_1 + q_2 + q_3)$$

subject to

$$\begin{aligned} p_1 - h_1 + q_1 &= 100, \\ p_2 + h_1 - q_1 - h_2 + q_2 &= 180, \\ p_3 + h_2 - q_2 - h_3 + q_3 &= 120, \\ h_3 - q_3 &= 0, & p_t, h_t, q_t, s_t &\geq 0, \\ p_1 + s_1 &= 150, \\ p_2 + s_2 &= 150, \\ p_3 + s_3 &= 150, \end{aligned}$$

There are

$$n = 12 \text{ variables}, \quad m = 7 \text{ equations.}$$

The variable order is $(p_1, p_2, p_3, h_1, h_2, h_3, q_1, q_2, q_3, s_1, s_2, s_3)$. Maximizing $-C$ gives the maximization convention if required.

If both h_t and q_t were positive, decreasing both by their minimum would preserve every constraint and decrease the objective. Consequently, at an optimum,

$$h_t = (I_t)^+, \quad q_t = (-I_t)^+.$$

In particular, the terminal equation gives $h_3 = q_3 = 0$ at an optimum. The LP therefore represents the original inventory/backlog cost.

Summing the three balance equations and using $I_0 = I_3 = 0$ gives

$$p_1 + p_2 + p_3 = 400.$$

Thus every feasible production plan has the same production cost, $3000(400) = 1200000$ DH.

Problem 5 (continued)

(c) For the proposed plan, take

$$p = (130, 150, 120), \quad h = (30, 0, 0), \quad q = (0, 0, 0), \quad s = (20, 0, 30).$$

All variables are nonnegative, and substitution verifies the seven equations:

$$130 - 30 = 100, \quad 150 + 30 = 180, \quad 120 = 120, \quad 0 = 0,$$

$$130 + 20 = 150, \quad 150 + 0 = 150, \quad 120 + 30 = 150.$$

Choose the seven basic variables

$$B = \{p_1, p_2, p_3, h_1, h_3, s_1, s_3\}.$$

The nonbasic variables are h_2, q_1, q_2, q_3, s_2 , all zero. To prove independence, consider the homogeneous system restricted to the columns of B . The terminal row and April-capacity row give

$$\Delta h_3 = 0, \quad \Delta p_2 = 0.$$

The April, March, and May balance rows then give successively

$$\Delta h_1 = 0, \quad \Delta p_1 = 0, \quad \Delta p_3 = 0.$$

The two remaining capacity rows give $\Delta s_1 = \Delta s_3 = 0$. The homogeneous system has only the zero solution, so these seven columns are independent (and the full matrix has row rank 7).

The plan is therefore a BFS. It is **degenerate**, because the basic variable h_3 is zero. Its cost is

$$C = 3000(130 + 150 + 120) + 200(30) = 1206000 \text{ DH.}$$

(d) With a late-delivery penalty of 100 DH per unit per month, postpone 30 April deliveries to May instead of storing them from March. This gives

$$p = (100, 150, 150), \quad I = (0, -30, 0).$$

The balances hold, production never exceeds 150, and all orders are settled by the end of May. There is no inventory cost and the only backlog is 30 units for one month. Hence

$$C = 3000(400) + 100(30) = 1203000 \text{ DH.}$$

To verify optimality, $p_2, p_3 \leq 150$ and total production 400 imply $p_1 \geq 100$, hence $I_1 \geq 0$. The April balance gives

$$I_1 - I_2 = 180 - p_2 \geq 30 \implies I_1 + (-I_2)^+ \geq 30.$$

The extra inventory/backlog cost of any feasible plan is therefore at least

$$200I_1 + 200(I_2)^+ + 100(-I_2)^+ \geq 100(I_1 + (-I_2)^+) \geq 3000.$$

The displayed plan attains this lower bound, proving optimality.

Problem 6

(a) The BFSs from Problem 1 project to

$$(20, 30), \quad (30, 0), \quad (0, 40), \quad (0, 0),$$

which are exactly the four vertices of the feasible polygon. Conversely, each vertex has a unique lift, since

$$s_1 = 80 - x_1 - 2x_2, \quad s_2 = 90 - 3x_1 - x_2.$$

Thus projection onto (x_1, x_2) is a bijection between the BFSs and the vertices.

A nonbasic variable is zero. If it is x_1 or x_2 , the point lies on the corresponding coordinate axis. If it is s_1 or s_2 , the corresponding resource inequality is tight. In this example, the two nonbasic variables identify the two boundary lines meeting at a vertex. In general, a basic variable can also be zero, so a zero coordinate need not be nonbasic.

(b) Three distinctions are necessary.

1. A selection of m columns may be dependent. It is then *not a basis* and does not define a unique basic solution.
2. An actual basis may produce a basic solution with a negative coordinate. It then contributes no BFS.
3. Different bases may produce the same BFS. This can occur when the solution is degenerate, since a zero variable can be basic. For example, in Problem 5, replacing h_3 by q_3 in the displayed basis gives the same point: the two columns are negatives of one another and both values are zero.

In Problems 1, 2, and 3, all six column pairs are independent and the feasible basic solutions are distinct. The respective counts 4, 3, and 4 arise solely because 2, 3, and 2 of the basic solutions have negative entries.

(c) Let $A \in \mathbb{R}^{m \times n}$ have row rank m and let $P \neq \emptyset$. Choose $x \in P$ whose support

$$S = \{j : x_j > 0\}$$

has minimum cardinality. Such a choice exists because support sizes are integers between 0 and n .

Suppose the columns A_S are dependent. Then there is a nonzero vector d , supported on S , with $Ad = 0$. Replacing d by $-d$ if necessary, assume $d_i > 0$ for at least one $i \in S$. Set

$$\alpha = \min_{i \in S: d_i > 0} \frac{x_i}{d_i} > 0, \quad x' = x - \alpha d.$$

We have $Ax' = Ax - \alpha Ad = b$. If $d_i > 0$, the definition of α ensures $x'_i \geq 0$; if $d_i \leq 0$, then $x'_i \geq x_i \geq 0$. Outside S , both x_i and d_i are zero. At least one positive coordinate becomes zero, and no new nonzero coordinate appears. Thus $x' \in P$ has strictly smaller support, a contradiction.

Therefore A_S is independent. Since $\text{rank}(A) = m$, extend S to a set B of m independent columns. The original x satisfies $x_j = 0$ for $j \notin B$, and $A_B x_B = b$. As A_B is invertible, this is its unique basic solution; it is nonnegative. Hence x is a BFS, proving that P contains at least one BFS. $\square$